SPM622 Software Engineering 2: Project & Dissertation

# TrackWIL

## Weekly Submission 4: Working Application, Chapters 4 and 5, Testing and Presentation Readiness

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Supervisor: Ms Amenda Raganya | Submitted: 23 October 2026

Table 1: Team details and links

| Item | Details |
|---|---|
| Project title | TrackWIL: Work-Integrated Learning Logbook Tracking |
| Group | Group 3, Cape Town Campus |
| Group lead | Willem (Gedion) Basson |
| GitHub repository | https://github.com/Noobey32/TrackWIL |
| Application access | Run locally from the repository: 05_Source_Code/trackwil (npm install, npm run dev); demonstration accounts are listed in the README |

## Contents of this Submission

Table 2: Week 4 requirements and where they are met

| Week 4 requirement | Where to find it |
|---|---|
| 1. Working application with the database integrated; walkthrough video | GitHub repository (05_Source_Code/trackwil, SQLite database); video link above |
| 2. Chapter 4 complete (4.6 to 4.8): development environment, modules and features with screenshots, version control | Chapter 4 (also in the dissertation draft) |
| 3. Chapter 5 draft: test strategy, test cases and results, defects, performance and usability evaluation, requirements evaluation, feedback | Chapter 5 (also in the dissertation draft) |
| 4. Testing workbooks | Test Plan; Test Cases and Results (81 test cases); Defect Log (7 defects); Requirements Traceability Matrix (25 requirements) |
| 5. Full dissertation draft: front matter and Chapters 1 to 5 with an outline of Chapter 6 | Dissertation draft (abstract, contents, lists of figures and tables, abbreviations, Chapters 1 to 5, Chapter 6 outline, references, appendices) |
| 6. Presentation plan and Individual Contribution Forms | Demo Script and Slide Outline; Individual Contribution Forms for all four members |
| Supporting evidence | Software Requirements Specification; Database Design document; Weekly Progress Report; Sprint Backlog; Risk, Issue and Change Register |

## Changes Since Week 3

Table 3: Changes made since the Week 3 submission

| Area | What changed | Where |
|---|---|---|
| Application | JSON file storage replaced with an SQLite relational database (keys, constraints, transactions); session cookie now signed; expired links can be resubmitted; mobile layout fixed; 36 automated system tests added | Pull request #5; branch gedion/fix-defects |
| Chapters 1 to 3 and SRS | Updated to match the system that was built (mentors verify by link without an account; coordinators and lecturers share one dashboard); scope split into included and excluded; measurable non-functional requirements; requirements-gathering evidence; Agile justified against Waterfall | Chapters 1 to 3; SRS |
| Chapter 2 | Technology review names the chosen stack; comparison table of similar systems; citations matched to the reference list | Chapter 2 |
| Chapter 4 | Completed (4.1 to 4.9); new class and activity diagrams; updated architecture, use case, sequence and ER diagrams; screenshots; version-control evidence | Chapter 4 |
| Testing | Test Plan rewritten; 55 test cases with actual results; 5 defects logged (3 fixed and retested); traceability matrix completed | Folder 06 workbooks; Chapter 5 |
| Formatting | Template covers, headers and footers removed; headings kept with their text, paragraphs not split across pages, figures and tables kept with their captions; one consistent heading structure | All documents |
| Project management | Sprint backlog, risk register and weekly progress report updated | Folder 09 |